Mateo Alado

Bay Area, CA | (510) 543-1444 | aladomateo@gmail.com | https://linkedin.com/in/mateoalado | https://github.com/matesuu

SKILLS

- **Languages:** Python, TypeScript, C/C++, SQL, Go, Java
- **AI/ML:** LLMs, RAG, GraphRAG, LangChain, LangGraph, PyTorch, Ollama, MCP
- **Backend:** FastAPI, REST APIs, GraphQL, Node.js, Cloudflare Workers, Distributed Systems
- **Database:** Vector Databases, PostgreSQL, MongoDB, Neo4j, Cloudflare D1
- **Tools:** Docker, AWS, Git

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern **Berkeley, CA**
Lawrence Berkeley National Laboratory June 2026 – Present

- Built a LangGraph-based agentic RAG system for the Advanced Light Source (ALS) as an internal platform to retrieve relevant information from 16,000+ ALS publications, providing reliable responses for domain-specific queries
- Reduced context retrieval and inference time from ~2 minutes to under 5 seconds by designing a custom PostgreSQL-backed GraphQL API for low-latency queries

Machine Learning Researcher **Hayward, CA**
California State University, East Bay January 2026 – June 2026

- Engineered and deployed a full-stack RAG research platform using FastAPI and Supabase for a team of 10 faculty researchers to query and manage 30+ academic papers and datasets
- Developed a backend inference engine using PolyG, reducing average token usage by 79.7% compared to similar open-source RAG frameworks

PROJECTS

FAIR2WISE - Autonomous Research Agents (Python, Academy, LangGraph, FastAPI, PostgreSQL, GraphQL)

- Engineered a pipeline of 3 HPC-enabled Academy agents to automate scientific literature ingestion, LinkML-based data extraction, and context retrieval for domain-specific RAG queries related to scientific research
- Implemented advanced features including source-code extraction, metadata ranking, and real-time graph UI rendering, expanding information access and usability for 450+ Berkeley Lab users (ALS and NERSC)

Multi-Model Coding Agent (TypeScript, Bun, Perplexity API, Cloudflare Workers, Cloudflare D1)

- Developed a terminal-native AI coding agent with multi-model orchestration (Codex, Claude Haiku/Sonnet/Opus), enabling dynamic routing of requests based on task complexity and token constraints
- Achieved an 88/100 on SWE benchmarks in multi-file debugging, async concurrency, and SQLAlchemy migrations, outperforming Codex (72) and Qwen 397B (58)

GraphRAG Research QA Platform (Python, TypeScript, React, OpenAI API, FastAPI, Neo4j, Supabase)

- Architected a RAG-based inference engine leveraging external knowledge graphs for adaptive graph traversal, optimizing long-context prompting and reducing model hallucinations
- Benchmarked against GraphRAG and reasoning-based baselines, achieving the highest F1 score (0.71) and 89% hit rate while reducing token usage by 75.5% compared to Microsoft GraphRAG and 93.3% compared to Graph-CoT

Visual Speech Recognition System (Python, PyTorch, NumPy, CNN, MediaPipe, OpenCV)

- Engineered a real-time lip-reading system and pipeline using MediaPipe Face Mesh and OpenCV webcam capture to classify speech patterns for hearing-impaired individuals
- Achieved a 94% validation accuracy across 24 speech classes through iterative CNN optimization and refinement

EDUCATION

California State University, East Bay **December 2026**
Bachelor of Science in Computer Science

- **Distinctions:** 4x Dean's Honor List, HackHayward 2026 Winner, Google Developer Student Club
- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Natural Language Processing, Software Engineering, Web Development, Linear Algebra